



Caringbah High School

2014

Trial HSC Examination

Mathematics Extension I

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen (Black pen is preferred)
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I Pages 2 – 4

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 5 – 8

60 marks

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Question 1 - 10 (1 mark each) Answer on page provided.

1) The remainder when $P(x) = 2x^3 + x^2 - 5x - 3$ is divided by $x + 1$ is:

- A) -3 B) -5 C) 1 D) -1

2) $\int \frac{1}{3+x^2} dx$ is given by:

- A) $\tan^{-1}\sqrt{3}x + c$ B) $\frac{1}{3}\tan^{-1}\frac{x}{3} + c$
C) $\frac{1}{3}\tan^{-1}3x + c$ D) $\frac{1}{\sqrt{3}}\tan^{-1}\frac{x}{\sqrt{3}} + c$

3) Given the function $f(x): y = \frac{2}{x+1}$, then its inverse function $f^{-1}(x)$ in terms of x is given by:

- A) $y = \frac{2-x}{x}$ B) $y = \frac{x+1}{2}$
C) $y = \frac{2-x}{2}$ D) $y = \frac{2+x}{x}$

4) $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$ is equal to:

- A) 1 B) $\frac{1}{3}$ C) -3 D) 3

5) The solution to the equation $3^{x-1} = 5$ is given by:

A) $x = \frac{\ln 5}{\ln 3} + 1$

B) $x = \frac{\ln 5}{\ln 3} - 1$

C) $x = \frac{\ln 3}{\ln 5} + 1$

D) $x = \frac{\ln 3}{\ln 5} - 1$

6) The general solution to the equation $2\sin \theta = \sqrt{3}$ is given by:

A) $\theta = k\pi + \frac{\pi}{3}$

B) $\theta = k\pi + (-1)^k \frac{\pi}{3}$

C) $\theta = k\pi + (-1)^k \frac{\pi}{6}$

D) $\theta = 2k\pi \pm \frac{\pi}{3}$

7) If two of the roots of $2x^3 - gx^2 + hx - 8 = 0$ are -1 and 2 , the other root is:

A) -2

B) 2

C) -4

D) 4

8) If $f(x) = \sin^2(3-x)$, then $f'(0)$ is equal to:

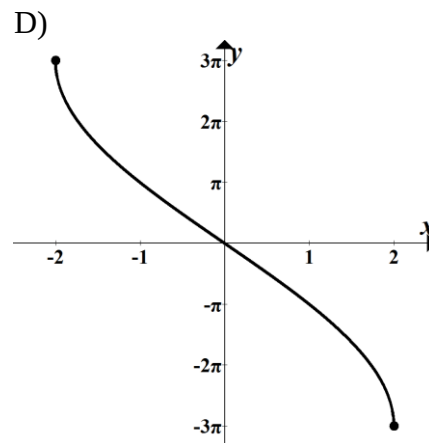
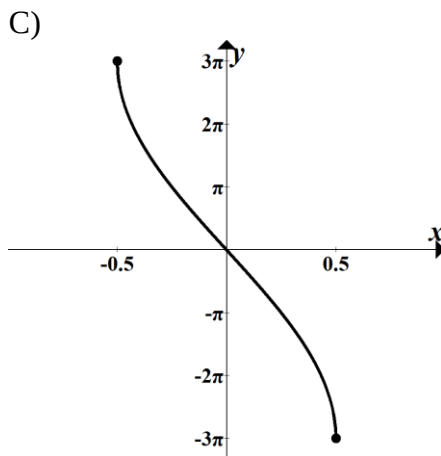
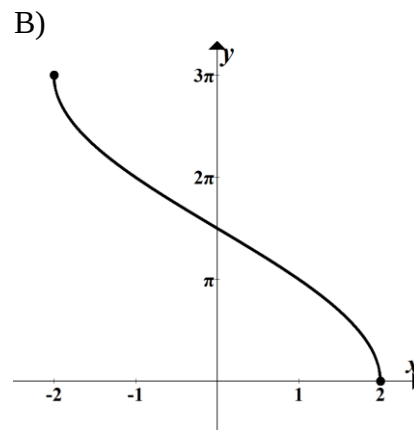
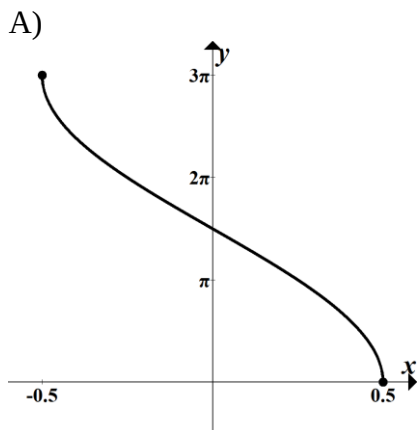
A) $-2\cos(3)$

B) $-2\sin(3)\cos(3)$

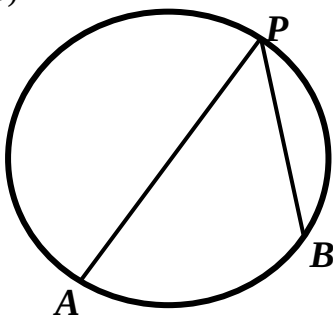
C) $2\sin(3)\cos(3)$

D) $6\sin(3)\cos(3)$

- 9) The graph of $y = 3\cos^{-1}\left(\frac{x}{2}\right)$ is:



10)



AP is a diameter of the circle.

If $\angle APB = 40^\circ$, then $\angle PAB$ is:

- A) 60° B) 40°
- C) 50° D) not enough information

END OF MULTIPLE CHOICE QUESTIONS

Section II**60 marks****Attempt all questions 11–14****Allow about 1 hour and 45 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a NEW booklet.**Marks**

- a) Solve $\frac{2x}{x-1} < 3$. 2
- b) Find the acute angle between the lines $x - 2y + 3 = 0$ and $3x + y + 6 = 0$. 2
[Answer to the **nearest degree**]
- c) Use the substitution $t = \tan \frac{\theta}{2}$ to express $1 + \tan \theta \tan \frac{\theta}{2}$ as a fraction 2
in simplest form.
- d) Find $\int \sin^2 x \, dx$. 2
- e) Given that a root of $y = 3x + \ln x - 1$ lies close to $x = 0.4$, use Newton's method 2
once to find an improved value of that root correct to 2 decimal places.
- f) Solve $\cos 2\theta = \cos \theta$ where $0 \leq \theta \leq 2\pi$. 2
- g) i) Write down the expansion of $\sin(A - B)$. 1
- ii) Hence show that $\sin 15^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}$. 2

Question 12 (15 marks) Start a NEW booklet.**Marks**

- a) The variable point $(3t, 4t^2)$ lies on a parabola. 2

Find the cartesian equation of this parabola.

- b) Find the domain of $y = \ln(\sin^{-1} x)$. 2

- c) Consider the function $f(x) = \frac{3x}{x^2 - 1}$.

- i) Show that the function is odd. 1

- ii) Show that the function is decreasing for all values of x . 2

- iii) Neatly sketch the graph of $f(x) = \frac{3x}{x^2 - 1}$ showing clearly the equations of any asymptotes. 2

- d) Use the substitution $u = x^2 - 2$ to evaluate $\int_{\sqrt{3}}^{\sqrt{11}} \frac{x}{\sqrt{x^2 - 2}} dx$ 3

- e) At time t hours after an oil spill occurs, a circular oil slick has a radius of ' r ' kilometres, where $r = \sqrt{t + 1} - 1$. 3

Find the rate at which the area of the slick is increasing when the radius is 1 kilometre.

Question 13 (15 marks) Start a NEW booklet.**Marks**

a) Find the exact value of $\cos\left(\sin^{-1}\left(\frac{3}{4}\right)\right)$. 1

b) i) Given that $A > 0$ and $0 < \alpha < \frac{\pi}{2}$, show that when $\cos x - \sin x$ 2

is expressed in the form $A\cos(x + \alpha)$, that $A = \sqrt{2}$ and $\alpha = \frac{\pi}{4}$.

ii) Hence solve $\cos x - \sin x = \frac{1}{\sqrt{2}}$ for $0 \leq x \leq 2\pi$. 2

c) A cake is cooling after being taken out of the oven. The surrounding temperature in the room is 20°C .

At a time ' t ' minutes, its temperature T decreases according to the equation

$$\frac{dT}{dt} = -k(T - 20) \text{ where 'k' is a positive constant. The initial temperature}$$

of the cake immediately after removal from the oven was 180°C and it cools to 120°C after 5 minutes.

i) Show that $T = 20 + Ae^{-kt}$ is a solution to $\frac{dT}{dt} = -k(T - 20)$, where 1

A is a constant.

ii) Find the values of A and k . 2

iii) How long will it take for the cake to cool to 60°C ? 2
(Answer to the nearest minute)

Question 13 continues on the next page

Question 13 continued.**Marks**

- d) The two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are on the parabola $x^2 = 4ay$.
- i) Show that the equation of the tangent at P is given by $y = px - ap^2$. 1
- ii) The tangents at P and Q meet at T . 2
Show that T has coordinates $[a(p + q), apq]$.
- iii) P and Q move in such a way that $\angle POQ$ is always 90° . (O is the origin). 1
Show that $pq = -4$.
- iv) Hence deduce the locus of T . 1

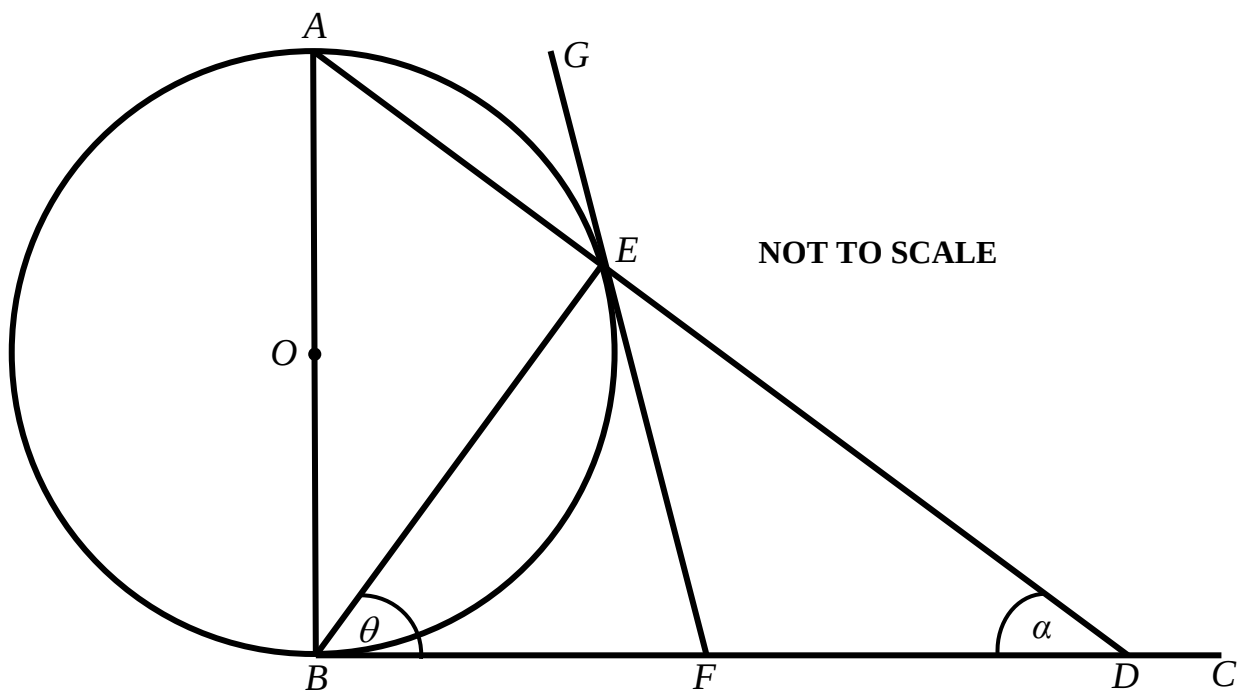
End of Question 13.

Question 14 (15 marks) Start a NEW booklet.**Marks**

- a) The polynomial $P(x) = 4x^3 + kx + 6$ has $(x+3)$ as a factor. 2

Find the value of k and express $P(x)$ in the form $(x+3)Q(x)$.

- b) In the diagram below, AB is the diameter of the circle centre O , and BC is tangential to the circle at B .
 The line AD intersects the circle at E and BC at D . The tangent to the circle at E intersects BC at F .
 Let $\angle EBF = \theta$ and $\angle EDF = \alpha$.



Answer this question on the page provided

- i) Prove that $\angle FED = \frac{\pi}{2} - \theta$ 2
- ii) Prove that $BF = FD$. 2

Question 14 continues on the next page

Question 14 continued.**Marks**

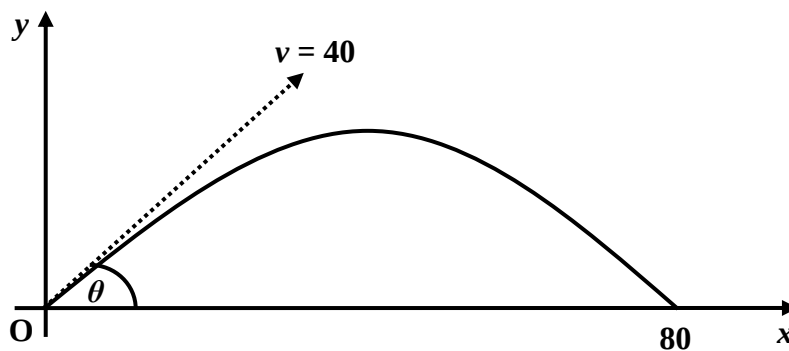
- c) Prove by Mathematical Induction that

3

$$1 \times 3 + 2 \times 3^2 + \dots + n \times 3^n = \frac{(2n-1)3^{n+1} + 3}{4}$$

where n is an integer, $n \geq 1$.

- d)



An arrow fired from ground level at a velocity of 40 m/s, is to strike the ground 80 metres away as shown in the diagram above.

- i) Show that the vertical and horizontal displacement equations are given respectively by: $y = 40t\sin\theta - 5t^2$ and $x = 40t\cos\theta$. (Assume $g = 10 \text{ m/s}^2$)
- ii) Hence, find the two angles at which the arrow can be fired.

2

4

END OF EXAM

Candidate Name/Number: _____

Multiple choice answer page. Fill in either A, B, C or D for questions 1-10.

This page must be handed in with your answer booklets

1.	
2.	
3.	
4.	
5.	

6.	
7.	
8.	
9.	
10.	

[illegible]

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Multiple Choice Section:

1.C 2.D 3.A 4.D 5.A
 6.B 7.A 8.B 9.B 10.C

Question 1.

The remainder is given by $P(-1)$

$$= 2(-1) + 1 - 5(-1) - 3 = 1 \quad \text{-----} \boxed{C}$$

Question 2.

$$\int \frac{1}{3+x^2} dx = \int \frac{1}{\sqrt{3}^2 + x^2} dx$$

$$\frac{1}{\sqrt{3}} \tan^{-1} \frac{x}{\sqrt{3}} + c \quad \text{-----} \boxed{D}$$

Question 3.

$$f(x): y = \frac{2}{x+1} \rightarrow f^{-1}(x): x = \frac{2}{y+1}$$

$$\therefore xy + x = 2 \rightarrow xy = 2 - x$$

$$\therefore y = \frac{2-x}{x} \quad \text{-----} \boxed{A}$$

Question 4.

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{x} = 3 \lim_{x \rightarrow 0} \frac{\sin 3x}{3x}$$

$$= 3 \times 1 = 3 \quad \text{-----} \boxed{D}$$

Question 5.

$$3^{x-1} = 5 \rightarrow x-1 = \log_3 5$$

$$\therefore x = 1 + \log_3 5$$

$$= 1 + \frac{\ln 5}{\ln 3} \quad \text{-----} \boxed{A}$$

Question 6.

$$\sin \theta = \frac{\sqrt{3}}{2} \rightarrow \text{Acute angle } \theta = \frac{\pi}{3}$$

$$\therefore \theta = k\pi + (-1)^k \frac{\pi}{3} \quad \text{-----} \boxed{B}$$

Question 7.

$$\text{product of roots} = \frac{-d}{a}$$

$$\therefore -1 \times 2 \times \alpha = \frac{-(-8)}{2} = 4$$

$$\therefore \alpha = -2 \quad \text{-----} \boxed{A}$$

Question 8.

$$f(x) = \sin^2(3-x) \rightarrow f'(x) = -2\sin(3-x)\cos(3-x)$$

$$\therefore f'(0) = -2\sin(3)\cos(3) \quad \text{-----} \boxed{B}$$

Question 9.

$$\text{Domain: } -1 \leq \frac{x}{2} \leq 1 \rightarrow -2 \leq x \leq 2$$

$$\text{Range: } 0 \leq y \leq 3\pi \quad \text{-----} \boxed{B}$$

Question 10.

$$\angle PBA = 90^\circ \{ \angle \text{ in semi-circle} \}$$

$$\text{Given } \angle APB = 40^\circ \rightarrow \angle PAB = 50^\circ \{ \angle \text{ sum } \triangle APB \}$$

$$\text{-----} \boxed{C}$$

Question 11.

$$\text{a) CV1: } 2x = 3(x-1) \rightarrow x = 3$$

$$\text{CV2: } x = 1$$

On testing $x < 1$ and $x > 3$, note $x \neq 1$

Question 11 continued.

$$\begin{aligned} \text{b) } x - 2y + 3 &= 0 \rightarrow m_1 = \frac{1}{2} \\ 3x + y + 6 &= 0 \rightarrow m_2 = -3 \end{aligned}$$

$$\therefore \tan \theta = \left| \frac{\frac{1}{2} - (-3)}{1 + \frac{1}{2} \times (-3)} \right| = 7$$

$$\therefore \theta = 81^\circ 52' \text{ (NOTE: accept } 82^\circ \text{)}$$

$$\text{c) } 1 + \tan \theta \tan \frac{\theta}{2} = 1 + \frac{2t}{1-t^2} \times t$$

$$= \frac{1-t^2+2t^2}{1-t^2}$$

$$= \frac{1+t^2}{1-t^2}$$

$$\text{d) } \cos 2x = 1 - 2\sin^2 x \rightarrow \sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\therefore I = \frac{1}{2} \int 1 - \cos 2x \, dx$$

$$= \frac{1}{2} \left(x - \frac{1}{2} \sin 2x \right) + c$$

$$= \frac{x}{2} - \frac{1}{4} \sin 2x + c$$

$$\text{e) Let } P(x) = 3x + \ln x - 1 \rightarrow P'(x) = 3 + \frac{1}{x}$$

$$a_1 = a_0 - \frac{P(a_0)}{P'(a_0)} \text{ (Newton's method)}$$

$$\boxed{P(0.4) = -0.7163; P'(0.4) = 5.5}$$

$$\begin{aligned} \therefore a_1 &= 0.4 - \frac{-0.7163}{5.5} \\ &\approx 0.53 \text{ (2 dp)} \end{aligned}$$

$$\text{f) } 2\cos^2 \theta - 1 = \cos \theta$$

$$\therefore 2\cos^2 \theta - \cos \theta - 1 = 0$$

$$(2\cos \theta + 1)(\cos \theta - 1) = 0$$

$$\therefore \cos \theta = -\frac{1}{2} \text{ and } \cos \theta = 1$$

$$\therefore \theta = \frac{2\pi}{3}, \frac{4\pi}{3}, 0, 2\pi$$

$$\text{g) i) } \sin(A-B) = \sin A \cos B - \sin B \cos A$$

$$\text{ii) } \sin(45^\circ - 30^\circ) = \sin 45^\circ \cos 30^\circ - \sin 30^\circ \cos 45^\circ$$

$$\therefore \sin 15^\circ = \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{3}-1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{6}-\sqrt{2}}{4}$$

$$\left[\text{Note: Could also use } \sin(60^\circ - 45^\circ) \right]$$

Question 12.

$$\text{a) } x = 3t \rightarrow t = \frac{x}{3}$$

$$\therefore y = 4t^2 \rightarrow y = 4\left(\frac{x}{3}\right)^2$$

$$\therefore y = \frac{4x^2}{9} \text{ or } 4x^2 = 9y$$

$$\text{b) For } \sin^{-1} x: \boxed{D: -1 \leq x \leq 1}$$

$$\text{and for } \ln(x): \boxed{D: x > 0}$$

$$\therefore \text{for } \ln(\sin^{-1} x): \boxed{D: 0 < x \leq 1}$$

$$\text{c) i) A function is odd if } f(-x) = -f(x)$$

$$f(x) = \frac{3x}{x^2-1} \rightarrow f(-x) = \frac{3(-x)}{(-x)^2-1}$$

$$= \frac{-3x}{x^2-1} = -f(x)$$

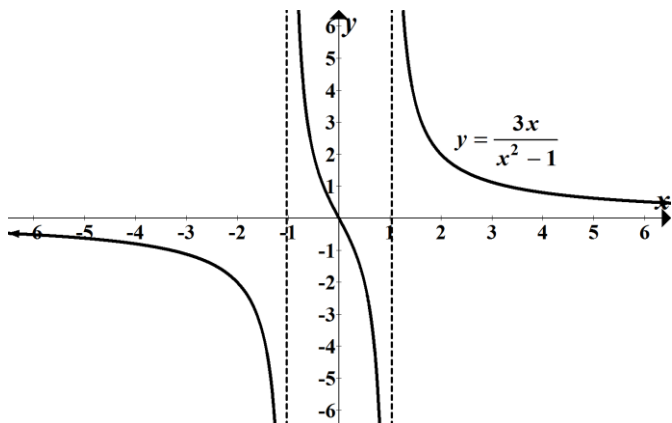
$$\text{ii) A function is decreasing if } f'(x) < 0 \text{ for all } x$$

$$f'(x) = \frac{(x^2-1) \times 3 - 3x \times 2x}{(x^2-1)^2}$$

$$= \frac{-3(x^2+1)}{(x^2-1)^2} < 0 \text{ for all } x$$

Question 12 continued:

- c) iii) Vertical asymptotes at $x = \pm 1$
Horizontal asymptote at $y = 0$ (x -axis)



d) $u = x^2 \rightarrow du = 2x dx$

When $x = \sqrt{3}, u = 1$; $x = \sqrt{11}, u = 9$

$$\therefore I = \frac{1}{2} \int_1^9 \frac{1}{\sqrt{u}} du = \frac{1}{2} \int_1^9 u^{-1/2} du$$

$$= \frac{1}{2} \left[2u^{1/2} \right]_1^9 = 3 - 1 = 2$$

e) $A = \pi r^2$; $r = \sqrt{t+1} - 1 \rightarrow r+1 = \sqrt{t+1}$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt} \text{ [chain rule]}$$

$$= 2\pi r \times \frac{1}{2}(t+1)^{-1/2}$$

$$= \pi r \times \frac{1}{\sqrt{t+1}}$$

$$\therefore \frac{dA}{dt} = \pi r \times \frac{1}{r+1}$$

and when $r = 1$:

$$\therefore \frac{dA}{dt} = \pi \times 1 \times \frac{1}{1+1} = \frac{\pi}{2} \text{ km/h}$$

Question 13.

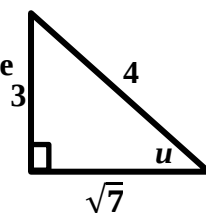
- a) Let $u = \sin^{-1}\left(\frac{3}{4}\right)$ and evaluate $\cos u$

$$u = \sin^{-1}\left(\frac{3}{4}\right) \rightarrow \sin u = \frac{3}{4}$$

$$\cos u = \sqrt{1 - \sin^2 u} \text{ or use a triangle}$$

$$= \sqrt{1 - \frac{9}{16}} = \sqrt{\frac{7}{16}}$$

$$\therefore \cos u = \frac{\sqrt{7}}{4} = \cos\left(\sin^{-1}\left(\frac{3}{4}\right)\right)$$



- b) i) Let $\cos x - \sin x = A \cos(x + \alpha)$
 $= A \cos \alpha \cos x - A \sin \alpha \sin x$

Equating coefficients of $\cos x$ and $\sin x$ gives:

$$A \cos \alpha = 1 \text{ --- [1] and } A \sin \alpha = 1 \text{ --- [2]}$$

$$[1]^2 + [2]^2 \rightarrow A^2 (\cos^2 x + \sin^2 x) = 2$$

$$\therefore A^2 = 2 \rightarrow A = \sqrt{2}$$

$$\text{Using [1] with } A = \sqrt{2} \rightarrow \cos \alpha = \frac{1}{\sqrt{2}}$$

$$\therefore \alpha = \frac{\pi}{4}$$

ii) $\cos x - \sin x = \frac{1}{\sqrt{2}}$

$$\therefore \sqrt{2} \cos\left(x + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \rightarrow \cos\left(x + \frac{\pi}{4}\right) = \frac{1}{2}$$

$$\therefore x + \frac{\pi}{4} = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\text{hence } x = \frac{\pi}{12}, \frac{17\pi}{12}$$

c) i) $T = 20 + Ae^{-kt} \rightarrow Ae^{-kt} = T - 20$

$$LHS = \frac{d(20 + Ae^{-kt})}{dt}$$

$$= -kAe^{-kt}$$

$$= -k(T - 20) = RHS$$

cii) when $t=0, T=180^\circ$; $t=5, T=120^\circ$

$$\therefore 180 = 20 + Ae^0 \rightarrow A = 160$$

$$\text{also } 120 = 20 + 160e^{-5k}$$

$$\therefore \frac{10}{16} = e^{-5k} \rightarrow -5k = \ln\left(\frac{5}{8}\right)$$

$$\therefore k = -\frac{1}{5}\ln\left(\frac{5}{8}\right) \rightarrow k \approx 0.094$$

$$\text{ii) } 60 = 20 + 160e^{-kt}$$

$$\therefore \frac{1}{4} = e^{-0.094t}$$

$$\therefore \ln\left(\frac{1}{4}\right) = -0.094t \rightarrow t \approx 15 \text{ minutes}$$

$$\text{d) i) } y = \frac{x^2}{4a} \rightarrow m = \frac{dy}{dx} = \frac{x}{2a}$$

$$\text{At } P: m = \frac{2ap}{2a} = p$$

$$\text{hence } y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$\therefore y = px - ap^2$$

$$\text{ii) } y = px - ap^2 \text{ ----- [1]}$$

$$\text{at } Q: y = qx - aq^2 \text{ ----- [2]}$$

$$[1] = [2] \rightarrow px - ap^2 = qx - aq^2$$

$$\therefore px - qx = ap^2 - aq^2$$

$$x(p - q) = a(p - q)(p + q)$$

$$\therefore x = a(p + q)$$

$$\text{hence } y = p \times a(p + q) - ap^2 = apq$$

$$\therefore T[a(p + q), apq]$$

$$\text{iii) since } \angle POQ = 90^\circ, m_{OP} \times m_{OQ} = -1$$

$$\therefore \frac{ap^2}{2ap} \times \frac{aq^2}{2aq} = -1$$

$$\therefore \frac{p}{2} \times \frac{q}{2} = -1 \rightarrow pq = -4$$

$$\text{iv) Since } pq = -4$$

$$y = apq \rightarrow y = -4a$$

which is the locus of T .

Question 14.

$$\text{a) } P(x) = 4x^3 + kx + 6$$

$$P(-3) = 0 \rightarrow -108 - 3k + 6 = 0$$

$$\therefore k = -34$$

$$\text{so } P(x) = 4x^3 - 34x + 6$$

$$\therefore 4x^3 - 34x + 6 = (x + 3)(4x^2 + bx + 2)$$

on equating coefficients of x^2 :

$$0 = b + 12 \rightarrow b = -12$$

$$\therefore P(x) = (x + 3)(4x^2 - 12x + 2)$$

Note:[or use long division]

$$\text{b) i) } \angle AEB = 90^\circ [\angle \text{ in semi-circle}]$$

$$BF = FE [\text{tangents from external point equal}]$$

$\therefore \triangle FBE$ is isosceles

$$\therefore \angle FEB = \theta [\angle \text{'s opposite equal sides in isos } \triangle]$$

and since $\angle BED = 90^\circ$, then

$$\angle FED = 90^\circ - \theta = \frac{\pi}{2} - \theta.$$

$$\text{ii) Using } \triangle ABD: \alpha = 180^\circ - \angle ABD - \angle BAD$$

$$= 180^\circ - 90^\circ - \theta$$

$$= 90^\circ - \theta \text{ or } \left(\frac{\pi}{2} - \theta\right)$$

$$\therefore \triangle FED \text{ is isosceles with } FE = FD$$

[sides opposite equal $\angle$'s in isos $\triangle$]

and also since $FE = BF$, then $BF = FD$.

$$\text{c) } 1 \times 3 + 2 \times 3^2 + \dots + n \times 3^n = \frac{(2n-1)3^{n+1} + 3}{4}$$

$$\text{When } n = 1: LHS = 1 \times 3 = 3$$

$$RHS = \frac{(2 \times 1 - 1) \times 3^2 + 3}{4}$$

$$= \frac{9 + 3}{4} = 3 \quad \therefore \text{true for } n=1.$$

Question 14 continued:

Assume true for $n = k$: i.e.

$$1 \times 3 + 2 \times 3^2 + \dots + k \times 3^k = \frac{(2k-1)3^{k+1} + 3}{4}$$

Prove true for $n = k$:

$$\text{i.e. } \boxed{S_k + T_{k+1} = S_{k+1}}$$

$$S_k = \frac{(2k-1)3^{k+1} + 3}{4}; S_{k+1} = \frac{(2k+1)3^{k+2} + 3}{4}; T_{k+1} = (k+1) \times 3^{k+1}$$

$$\begin{aligned} LHS &= \frac{(2k-1)3^{k+1} + 3}{4} + (k+1) \times 3^{k+1} \\ &= \frac{(2k-1)3^{k+1} + 3 + 4(k+1) \times 3^{k+1}}{4} \\ &= \frac{3^{k+1}(2k-1 + 4(k+1)) + 3}{4} \\ &= \frac{3^{k+1}(6k+3) + 3}{4} \\ &= \frac{(2k+1) \times 3 \times 3^{k+1} + 3}{4} \\ &= \frac{(2k+1) \times 3^{k+2} + 3}{4} \\ &= S_{k+1} = RHS \end{aligned}$$

$\therefore$ If true for $n = k$, then true for $n = k + 1$.

Hence by the principle of mathematical induction,
the result is true for all $n \geq 1$.

d) i) $t = 0, y = 0, \dot{y} = 40 \sin \theta, x = 0, \dot{x} = 40 \cos \theta$

Vertically: $\ddot{y} = -10 \rightarrow \dot{y} = -10t + c$

$t = 0, \dot{y} = 40 \sin \theta \rightarrow c = 40 \sin \theta$

$\therefore \dot{y} = 40 \sin \theta - 10t$

$\therefore y = 40t \sin \theta - 5t^2 + c$

$0 = 40 \sin \theta \times 0 - 5 \times 0 + c \rightarrow c = 0$

$\therefore y = 40t \sin \theta - 5t^2$

Horizontally: $\ddot{x} = 0 \rightarrow \dot{x} = 40 \cos \theta$

$\therefore x = 40t \cos \theta + c$

$0 = 40 \cos \theta \times 0 + c \rightarrow c = 0$

$\therefore x = 40t \cos \theta$

ii) When $y = 0, 0 = t(40 \sin \theta - 5t)$

$\therefore$ when it strikes the ground $t = 8 \sin \theta$

$\therefore$ when $t = 8 \sin \theta, x = 80$

$\therefore 80 = 40 \cos \theta \times 8 \sin \theta$

$\frac{1}{2} = 2 \sin \theta \cos \theta$

$\frac{1}{2} = \sin 2\theta$

$\therefore 2\theta = 30^\circ, 180^\circ - 30^\circ$

hence the two angles are 15° and 75° .